

Problem 1

- a) Need to label two feed streams, one outlet stream, and %xs air condition
- b) DoF = 1, unless basis is already assumed, then 0
- c) Answer depends on whether or not a basis was already assumed
- d) Balances will depend on method chosen (EoR or atomic)

Problem 2

- a) Answer given
- b) Plot y vs. $\sqrt{x}$, slope = a , intercept = b
- c) Plot $\ln(q)$ vs. g , slope = b , intercept = $\ln(a)$
- d) Plot w^3 vs. t^2 , slope = a , intercept = $-b$
- e) Plot $\sqrt{\frac{x}{\sin(y)}}$ vs. x , slope = a , intercept = b
- f) Plot $\ln(p)$ vs. $1/T$, slope = $-E/k$, intercept = $\ln(A)$
- g) Plot $\ln(\Gamma)$ vs. $\ln(\gamma)$, slope = n , intercept = $\ln(k)$

Problem 3

- a) $C2 = B\$16 * A2 / (B\$15 + A2)$
 $D2 = (B2 - C2)^2$
- b) $D15 = \text{sum}(D2:D13)$
- c) Set Objective: D15
 Goal: min, or value of 0
 By Changing Variable Cells: B15 and B16
 Hit button: Solve

Problem 4

- a) Unit of a : $\text{atm} \cdot \text{L}^2 / \text{mol}^2$
 Unit of b : L/mol
- b) Plot p vs. T , slope = $\frac{R}{(\hat{V}-b)}$, intercept = $-a\left(\frac{1}{\hat{V}}\right)^2$
- c)
$$p = \left\{ \frac{R}{(\hat{V}-b)} \left[\frac{T(^{\circ}\text{F})-32}{1.8} + 273.15 \right] - a\left(\frac{1}{\hat{V}}\right)^2 \right\} \frac{1.01325 \times 10^5 \text{ Pa}}{1 \text{ atm}}$$